

Practice Quiz: Digital Transformation — Module 03:

Perception

Name: ____ Date: ____

Part A: Multiple Choice

1. **Analytics maturity** progresses from: A) Cheap to expensive B) Descriptive (what happened) → diagnostic (why) → predictive (what will happen) → prescriptive (what to do) C) Simple to complex D) Fast to slow
 2. Target's pregnancy prediction illustrates: A) Good marketing B) The data-to-decision gap — accurate analytics doesn't automatically produce good organizational action C) Privacy violation D) Technology superiority
 3. **Data quality** is fundamental because: A) Clean data looks nice B) Errors in perception propagate through all downstream inference — garbage in, garbage out C) Regulators require it D) Data is expensive
 4. **Real-time sensing** changes organizations by: A) Making them nervous B) Enabling faster inference-action cycles that can respond to the environment as it changes C) Creating more work D) Requiring more servers
 5. The best analytics investment for most organizations is: A) The most advanced AI B) Improving data quality and descriptive analytics before building predictive C) More dashboards D) Hiring data scientists
 6. **Data bias** distorts organizational perception by: A) Making data biased B) Producing systematically skewed inferences that misrepresent reality C) Being unethical D) Affecting predictions
 7. A/B testing is Active Inference because: A) It uses two groups B) It systematically tests a hypothesis by observing outcomes under different conditions, updating the organization's model C) It's experimental D) It's digital
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Part B: Short Analysis

1. Your organization has 47 dashboards but executives say they "can't find the data they need." Diagnose this problem using the data-to-decision gap concept and design a solution.
2. Design a real-time sensing system for a retail organization. Specify: what data to collect, how to process it, what decisions it informs, and how frequently.
3. Your analytics team discovers that the training data for a customer churn model systematically underrepresents customers from certain demographics. Analyze the implications and design a remediation strategy.